

Homework #3

Logic

1. Let $A = \{1, 3, 12, 35\}$, $B = \{3, 7, 12, 20\}$, and $C = \{x \mid x \text{ is a prime number}\}$. List the elements of the following sets. Are any of the sets below disjoint from any of the others? Are any of the sets below subsets of any others?

(a) $A \cap B$

$\{3, 12\}$

(b) $(A \cup B) \setminus C$

$\{1, 3, 7, 12, 20, 35\}$

(c) $A \cup (B \setminus C)$

$\{1, 3, 12, 20, 35\}$

2. Use any method you wish to verify the following identities:

(a) $(A \setminus B) \cap C = (A \cap C) \setminus B$

Conditions for membership on the LHS:

$$(A \setminus B) \cap C \implies (x \in A \wedge x \notin B) \wedge (x \in C)$$

Conditions for membership on the RHS:

$$(A \cap C) \setminus B \implies (x \in A \wedge x \in C) \wedge (x \notin B)$$

The and operator is associative, therefore the expressions are equivalent.

(b) $(A \cap B) \setminus B = \emptyset$

Conditions for membership on the LHS:

$$(A \cap B) \setminus B \implies (x \in A \wedge x \in B) \wedge x \notin B$$

$x \in B \wedge x \notin B$ is a contradiction, there is no element that is in B and not in B, so it equals the empty set

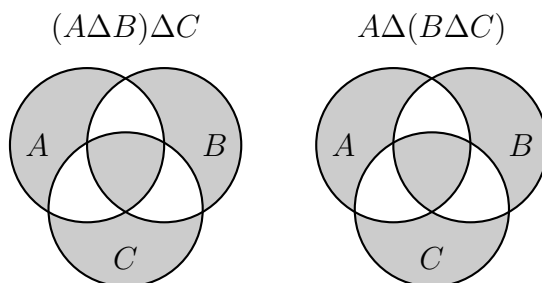
(c) $(A \setminus B) \setminus B = (A \cap B)$

Conditions for membership on the LHS:

$$(A \setminus B) \setminus B \implies (x \in A \wedge x \notin B) \wedge x \notin B$$

The last one is redundant and already implies that any element must be in A and not in B, which is the definition of a set difference.

3. Use a Venn Diagram to show the associativity of the Symetric Difference



This makes sense, as the symmetric difference operator essentially acts like the XOR, so only items that are not in either set get left. And we also know that XOR is associative as well, so this holds up.

4. Analyze the logical forms of the following statements:

(a) Mary will sell her house only if she can get a good price and find a nice apartment.

P = Mary will sell her house

Q = She gets a good price

R = She finds a nice apartment

$P \implies Q \wedge R$

(b) Having both a good credit history and an adequate down payment is a necessary condition for getting a mortgage.

P = Getting a mortgage

Q = Having a good Credit history

R = Having an adequate down payment

$P \implies Q \wedge R$

(c) John will drop out of school, unless someone stops him. (Hint: First try to rephrase this using the words if and then instead of unless.)

P = John Drops out

Q = Someone stops him

$\neg P \implies Q$

(d) If x is divisible by either 4 or 6, then it isn't prime.

P = x is Prime

$Q = x \mid 4$

$R = x \mid 6$

$Q \vee R \implies \neg P$

5. Show that $(P \rightarrow Q) \wedge (Q \rightarrow R)$ is equivalent to $(P \rightarrow R) \wedge [(P \iff Q) \vee (R \iff Q)]$.

Table 1: Truth Table Verification of Logical Equivalence

P	Q	R	$P \rightarrow Q$	$Q \rightarrow R$	LHS	$P \rightarrow R$	$P \iff Q$	$R \iff Q$	RHS
T	T	T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T	F	F
T	F	T	F	T	F	T	F	F	F
T	F	F	F	T	F	F	F	T	F
F	T	T	T	T	T	T	F	T	T
F	T	F	T	F	F	T	F	F	F
F	F	T	T	T	T	T	T	F	T
F	F	F	T	T	T	T	T	T	T

6. Which of the following formulas are equivalent?

(a) $P \rightarrow (Q \rightarrow R) \equiv \neg P \vee \neg Q \vee R$

(b) $Q \rightarrow (P \rightarrow R) \equiv \neg Q \vee \neg P \vee R \equiv \neg P \vee \neg Q \vee R$

(c) $(P \rightarrow Q) \vee (P \rightarrow R) \equiv (\neg P \vee Q) \vee (\neg P \vee R) \equiv \neg P \vee \neg Q \vee R$

(d) $(P \wedge Q) \rightarrow R \equiv \neg(P \wedge Q) \vee R \equiv \neg P \vee \neg Q \vee R$

(e) $P \rightarrow (Q \wedge R) \equiv \neg P \vee (Q \wedge R) \equiv (\neg P \vee Q) \wedge (\neg P \vee R)$

(a), (b), (c), and (d) are all logically equivalent to each other.